

SMART INDIA HACKATHON 2026



- **Problem Statement ID – 26103**
- **Problem Statement Title- Use case on web-based integrated project-monitoring platform**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-**
- **Team Name – Tech Panther**



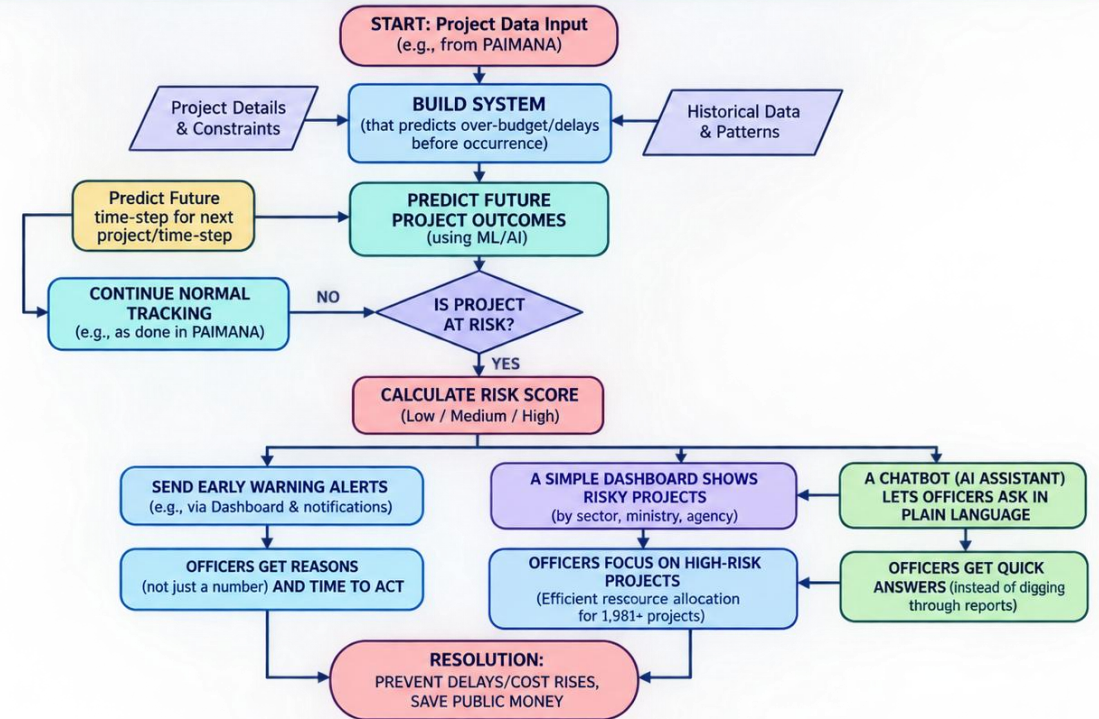
❑ Idea/Solution

- ❖ Build a system that predicts which projects will go over budget or get delayed, before it actually happens..
- ❖ Give every project a **risk score (Low / Medium / High)** so officers know where to look first.
- ❖ Send **early warning alerts** when a project starts showing warning signs, like **slow progress or rising cost**.
- ❖ A simple dashboard shows **risky projects by sector, ministry and agency**.
- ❖ A **chatbot (AI assistant)** lets officers ask in plain language, like "Which road projects are likely to be delayed?"

❑ Problem Resolution

- ❖ Our system tells what will likely happen.
- ❖ Predicts what is likely to happen.
- ❖ Provides early alerts before costs rise.
- ❖ Prioritizes high-risk projects for officers.
- ❖ Explains the reasons behind delays and cost increases.
- ❖ Provides quick answers through a chatbot.
- ❖ Saves public money and delivers assets sooner.

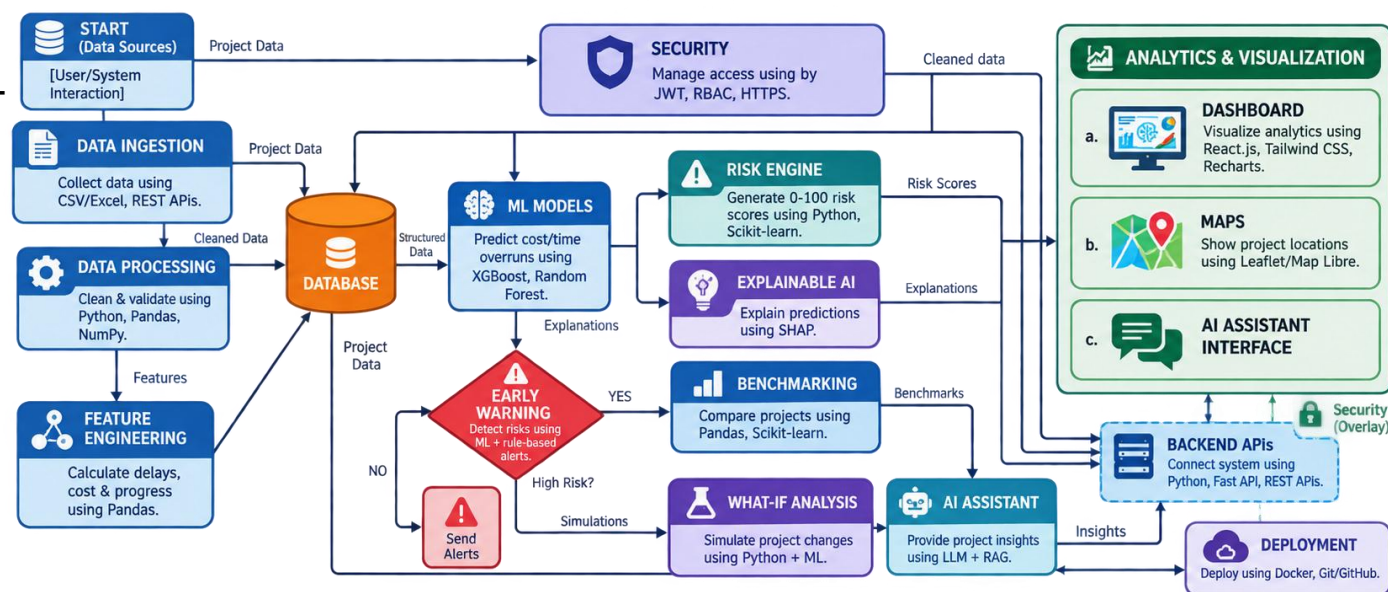
PREDICTIVE RISK SYSTEM FOR INFRASTRUCTURE PROJECTS: IDEA & RESOLUTION FLOWCHART (Derived from 'TECH PANTHER' idea for SMART INDIA HACKATHON 2026)



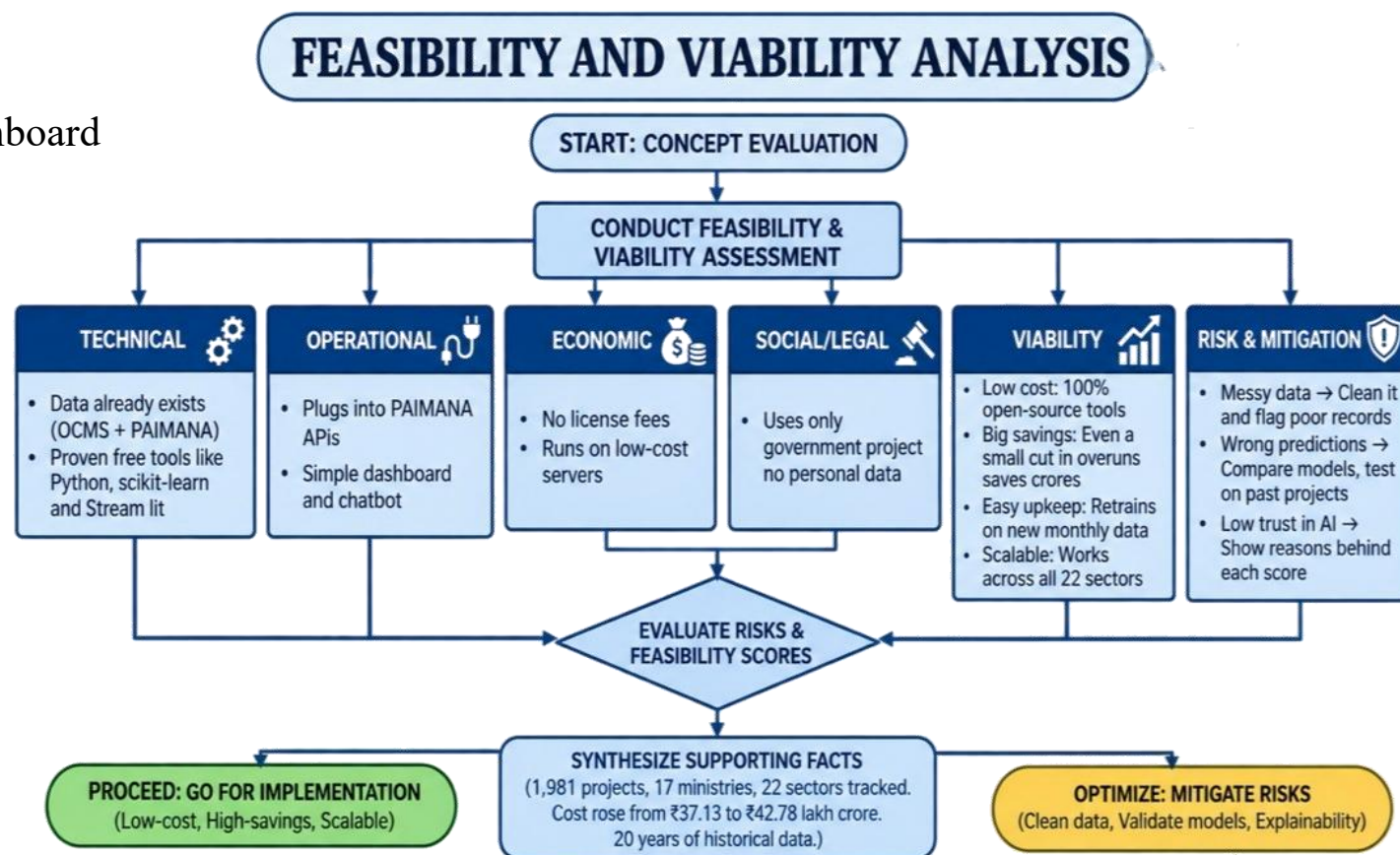
❑ Unique Value Propositions (UVP)

- ❖ **Predicts, not just reports:** Identifies future risks early.
- ❖ **100% Open-Source:** Low cost with no vendor lock-in.
- ❖ **Explainable AI:** Shows the key reasons behind each risk score.
- ❖ **Proven AI Value:** Compares AI with traditional statistical methods.
- ❖ **Identifies Data Gaps:** Suggests missing fields for CUF improvement.
- ❖ **Easy to Use:** Simple chatbot and dashboard for all officers.
- ❖ **PAIMANA Integration:** Uses monthly data and APIs for continuous learning.

- ❑ Data Ingestion – Collect data using CSV/Excel, REST APIs.
- ❑ Data Processing – Clean & validate using Python, Pandas, NumPy.
- ❑ Feature Engineering – Calculate delays, cost & progress using Pandas.
- ❑ Database – Store project data in PostgreSQL, SQL.
- ❑ ML Models – Predict cost/time overruns using XG Boost, Random Forest.
- ❑ Risk Engine – Generate 0–100 risk scores using Python, Scikit-learn.
- ❑ Explainable AI – Explain predictions using SHAP.
- ❑ Early Warning – Detect risks using ML + rule-based alerts.
- ❑ Benchmarking – Compare projects using Pandas, Scikit-learn.
- ❑ What-If Analysis – Simulate project changes using Python + ML.
- ❑ AI Assistant – Provide project insights using LLM + RAG.
- ❑ Dashboard – Visualize analytics using React.js, Tailwind CSS, Recharts.
- ❑ Maps – Show project locations using Leaflet/Map Libre.
- ❑ Backend APIs – Connect system using Python, Fast API, REST APIs.
- ❑ Security – Manage access using JWT, RBAC, HTTPS.
- ❑ Deployment – Deploy using Docker, Git/GitHub.

TECHNICAL APPROACH - SYSTEM
ARCHITECTURE FLOWCHART

- ❑ **Technical:** Data already exists (OCMS + PAIMANA). Proven free tools like Python, scikit-learn and Stream lit.
- ❑ **Operational:** Plugs into PAIMANA APIs. Simple dashboard and chatbot.
- ❑ **Economic:** No license fees. Runs on low-cost servers.
- ❑ **Social/Legal:** Uses only government project data, no personal data.
- ❑ **Viability**
Low cost: 100% open-source tools.
Big savings: Even a small cut in overruns saves crores.
Easy upkeep: Retrains on new monthly data.
Scalable: Works across all 22 sectors.
- ❑ **Risk and Mitigation**
Messy data → Clean it and flag poor records.
Wrong predictions → Compare models, test on past projects.
Low trust in AI → Show reasons behind each score.
- ❑ **Supporting Facts**
1,981 projects, 17 ministries, 22 sectors tracked.
Cost rose from ₹37.13 to ₹42.78 lakh crore (about 15% extra).
20 years of historical data ready to use.





Government / Monitoring Impact

- Problems detected early, avoiding cost and delay escalation
- Officers prioritize projects effectively with limited staff/time
- Data-driven decisions replace guesswork
- Monitoring shifts from reactive reporting to proactive prevention



Economic Benefits

- Saves public money by reducing cost overruns on mega projects
- Faster completion reduces cost escalation and accelerates returns
- Low operating cost due to open-source tools
- Improves future budget and fund planning



Social Benefits

- Citizens receive infrastructure (roads, railways, water, power) sooner
- Timely project completion reduces community disruption
- Greater transparency and accountability in public projects



Benefits for Ministries & Agencies

- Early alerts help resolve land, contractor, or approval issues
- Benchmarking enables cross-sector project comparisons
- Clear reasons for delays/cost rise prevent repeat mistakes



Benefits for MoSPI

- Identifies new CUF fields for better predictions
- Utilizes 20 years of OCMS and PAIMANA data
- Tests AI vs. traditional statistics for future tech decisions



Environmental Benefits

- Reduced delays minimize wasted materials, idle machinery, and rework
- Faster delivery of clean energy, water, and transport projects



Scalability

- Applicable across 22 sectors and 17 ministries in PAIMANA
- Extendable to state-level and smaller projects (< ₹150 crore)
- Improves continuously with new monthly data
- Reusable by other departments tracking large projects

Project Benefits Overview



Government & Monitoring

- Early Issue Detection
- Smart Project Focus
- Proactive Management



Economic Impact

- Cost Savings
- Faster Completion
- Efficient Spending



Social Improvement

- Faster Infrastructure
- Reduced Disruption
- Greater Transparency



Ministries & Agencies

- Early Alerts
- Performance Comparison
- Root Cause Insights



MoSPI Advancements

- New Data Insights
- 20 Years of Analysis
- AI & Innovation



Environmental & Scalability

- Less Waste
- Green Projects
- Broad Adaptability

- **PAIMANA / OCMS** : Studied existing Government project-monitoring system and available project data.
- **Data Analysis**: Reviewed project cost, progress, expenditure, delay and completion parameters.
- **Research Gap**: Existing monitoring mainly reports current/past status; our solution adds predictive risk analysis.
- **ML Research**: Random Forest & XG Boost studied for predicting cost and time overruns.
- **Explainable AI**: SHAP used to identify the major factors behind each prediction.
- **Risk & Alerts**: Research supports combining ML predictions + rule-based thresholds for early warnings.
- **Decision Support**: Benchmarking, What-if analysis and AI/RAG assistant convert predictions into actionable insights.
- **Technology Research**: Python, Pandas, Scikit-learn, PostgreSQL, Fast API and React/Tailwind selected based on suitability and scalability.
- **References** : MoSPI – PAIMANA / IPMDOCMS Project Monitoring PAIMANA Public Dashboard Research papers on ML-based project delay & cost prediction Research on SHAP / Explainable AI

Existing System vs Our Platform

From Monitoring the Past to Predicting the Future

Feature	 Existing System (PAIMANA / OCMS)	 Our Platform (Tech Panther)
 Project Monitoring	Current project status	Current status + future risk prediction
 Cost Tracking	Original & revised cost, expenditure	Cost tracking + overrun prediction
 Delay Monitoring	Tracks time overrun (after it happens)	Predicts delay before it occurs
 Risk Score	Not available	0 – 100 Risk Score (Low / Medium / High)
 Reason for Risk	Issues / slippages reported	AI/ML + SHAP explanations (reasons)
 Project Comparison	Sector-wise reports	AI-based benchmarking (across sectors)
 What-if Analysis	Not available	Simulate project changes (time, cost, scope)
 AI Assistant	Not available	LLM + RAG chatbot for quick answers
 Dashboard	Reports and dashboard	Predictive risk dashboard with insights
 Maps	Project location information	Interactive project maps (Leaflet / MapLibre)
 Decision Support	Monitoring and corrective action	Predictive + actionable insights for officers